Year 2 Pure Mathematics

Chapter 10

Numerical methods

Practice Booklet

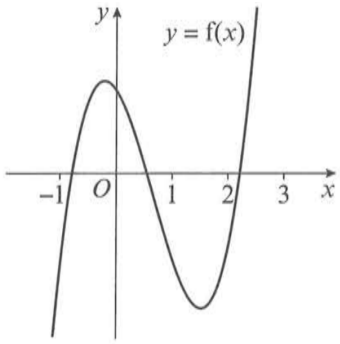
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10.1 Locating roots

Practice Book, printed page 81.

1. The diagram shows a sketch of the curve $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = 2x^3 - 4x^2 - 2x + 2$.



Hint

If the function $\mathrm{f}(x)$ is continuous on the interval $[a, b]$ and $\mathrm{f}(a)$ and $\mathrm{f}(b)$ have opposite signs, then $\mathrm{f}(x)$ has at least one root, x, which satisfies $a < x < b$.

a. Explain how the graph shows that $\mathrm{f}(x)$ has a root between $x = -1$ and $x = 0$.

b. Calculate:

i. $\mathrm{f}(2.2)$

ii. $\mathrm{f}(2.3)$

c. Explain how your answer to part b shows that $\mathrm{f}(x)$ has a root between $x = 2.2$ and $x = 2.3$.

2. Show that each of these functions has at least one root in the given interval.

a. $\mathrm{f}(x) = 4 - \frac{1}{2}x^2 + 2x^3$, $-1.2 < x < -1.1$

b. $\mathrm{f}(x) = x^2 \ln x - 2e^x + 4$, $0.6 < x < 0.7$

c. $\mathrm{f}(x) = 3x^2 + \frac{5}{x^3}$, $-1.11 < x < -1.10$

d. $\mathrm{f}(x) = 9 + \frac{3}{4}x - 7\sqrt{x}$, $2.37 < x < 2.38$

Hint

Evaluate $\mathrm{f}(x)$ at the endpoints of the given interval and show that $\mathrm{f}(x)$ changes sign in that interval.

3.

a. On the same set of axes, sketch the graphs of $y = \ln x$ and $y = -(x + 1)^2 + 6$.

b. Explain why the function $\mathrm{f}(x) = \ln x + (x + 1)^2 - 6$ has only one root.

c. Show that the function $\mathrm{f}(x) = \ln x + (x + 1)^2 - 6$ has a root between $x = 1.3$ and $x = 1.4$.

Hint

If $\ln x + (x+1)^2 - 6 = 0$, then $\ln x = -(x+1)^2 + 6$.

4. $f(x) = \cos^2 x - e^{-2x}$, where x is in radians.

a. Find $f'(x)$.

b. Show that there is a stationary point, α, of $y = f(x)$ in the interval $[3.1, 3.2]$.

c. By considering a change in sign of $f'(x)$ in a suitable interval, verify that $\alpha = 3.143$ to 3 decimal places.

Hint

A stationary point occurs where the gradient is zero, so look for a sign change in the gradient.
⟵ Year 1, Section 12.9

Hint

To determine a root to a given degree of accuracy you need to show it lies within the range of values that all round to the given value. Here consider $f'(3.1425)$ and $f'(3.1435)$.

5. E $g(x) = \frac{1}{2}x^3 - 2\cos x$, where x is in radians.

a. Show that $g(x)$ has a root, α, in the interval $1.1 < x < 1.2$. (2 marks)

b. By considering a change of sign of $g(x)$ in a suitable interval, verify that $\alpha = 1.165$ correct to 3 decimal places. (3 marks)

6. E/P

a. On the same set of axes, sketch the graphs of $y = (x+2)^2 - 1$ and $y = 2^x$. (2 marks)

b. With reference to your sketch explain why $f(x) = 2^x + 1 - (x+2)^2$ has two roots. (2 marks)

c. Show that the function $f(x) = 2^x + 1 - (x+2)^2$ has a root in the interval $-0.8 < x < -0.7$. (3 marks)

7. E/P $f(x) = \dfrac{2}{6-3x} - 4$

a. Calculate $f(1.9)$ and $f(2.1)$. (2 marks)

b. Explain why the equation $f(x) = 0$ does not have a root in the interval $1.9 < x < 2.1$. (2 marks)

The equation $f(x) = 0$ has a single root α.

c. Use algebra to find the exact value of α. (2 marks)

8. E/P

a. On the same set of axes, sketch the graphs of $y = \frac{1}{4}\sin 2x$ and $y = \ln x$. (2 marks)

b. With reference to your sketch explain why $h(x) = \sin 2x - 4\ln x$ has only one root. (2 marks)

c. Show that the function $h(x) = \sin 2x - 4\ln x$ has a root in the interval $[1.18, 1.19]$. (3 marks)

10.2 Iteration

Practice Book, printed page 82.

1. $f(x) = x^2 - 8x + 3$

Show that $f(x) = 0$ can be written as:

a. $x = \sqrt{8x - 3}$

b. $x = 8 - \dfrac{3}{x}$, $x \neq 0$

c. $x = \dfrac{x^2 - 3}{8}$

Hint

Rearrange $x^2 - 8x + 3 = 0$ to obtain each answer.

2. $f(x) = x^2 - 4x - 2$

a. Show that $f(x) = 0$ can be written as $x = \sqrt{4x + 2}$.

b. Use the iterative formula $x_{n+1} = \sqrt{4x_n + 2}$, $x_0 = 5$, to find, to 3 decimal places, the values of x_1, x_2, x_3 and x_4.

Hint

Use your calculator to find the values of x_1, x_2, x_3 and x_4 efficiently.

3. $h(x) = -x^3 + 3x^2 - 1$

a. Show that $h(x) = 0$ can be written as:

i. $x = \sqrt[3]{3x^2 - 1}$

ii. $x = \sqrt{\dfrac{x^3 + 1}{3}}$

b. Use the iterative formula $x_{n+1} = \sqrt[3]{3x_n^2 - 1}$, $x_0 = 1$, to find, to 3 decimal places, the values of x_1, x_2, x_3 and x_4.

c. Use the iterative formula $x_{n+1} = \sqrt{\dfrac{x_n^3 + 1}{3}}$, $x_0 = 1$, to find, to 3 decimal places, the values of x_1, x_2, x_3 and x_4.

d. Use your answers to parts b and c to explain why there could be more than one solution to the equation $h(x) = 0$.

4. E/P $f(x) = 2\sin x - \dfrac{1}{2}x^2$

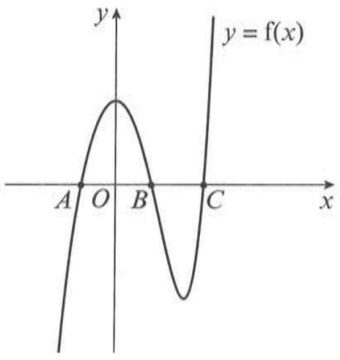
a. Show that the equation $f(x) = 0$ can be written as $x = \sqrt{4\sin x}$. (2 marks)

b. Let $x_0 = 1.5$. Use the iterative formula $x_{n+1} = \sqrt{4\sin x_n}$ to find, to 3 decimal places, the values of x_1, x_2 and x_3. (3 marks)

The root of $f(x)$ is α.

c. By considering the change of sign of $f(x)$ in a suitable interval, prove that $\alpha = 1.934$ to 3 decimal places. (3 marks)

5. E/P The diagram shows a sketch of the curve $y = f(x)$, where $f(x) = x^5 - 5x^2 + 2$.



The x-coordinates of the points A and C are also solutions to the equation $x^2 - x - 1 = 0$.

a. Find the exact coordinates of the points A and C. (3 marks)

b. Show that the equation $f(x) = 0$ can be written as $x = \sqrt{\dfrac{x^5 + 2}{5}}$ (2 marks)

c. Let $x_0 = 0.5$. Use the iterative formula $x_{n+1} = \sqrt{\dfrac{x_n^5 + 2}{5}}$ to find, to 3 decimal places, the values of x_1, x_2 and x_3. (3 marks)

The root of $f(x)$ at point B is α.

d. By considering a suitable interval, prove that $\alpha = 0.651$ to 3 decimal places. (3 marks)

6. E/P $\mathrm{f}(x) = e^{x+4} + \frac{1}{2}x^2 - 10$

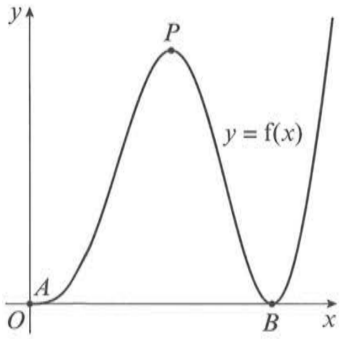
a. Show that there is a root α of $\mathrm{f}(x)$ in the interval $[-1.9, -1.8]$. (2 marks)

b. Show that the equation $\mathrm{f}(x) = 0$ can be written as $x = \ln\left(10 - \frac{1}{2}x^2\right) - 4$. (2 marks)

c. Let $x_0 = -3$. Use the iterative formula $x_{n+1} = \ln\left(10 - \frac{1}{2}x_n^2\right) - 4$ to find, to 3 decimal places, the values of x_1, x_2 and x_3. (3 marks)

d. By considering a suitable interval, prove that $\alpha = -1.8954$ to 4 decimal places. (3 marks)

7. E/P The diagram shows the part of the graph of $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = x\sin^2 x$. The curve touches the x-axis at A and B and has a turning point at P.



a. Verify that the coordinates of A and B are $(0, 0)$ and $(\pi, 0)$ respectively. (1 mark)

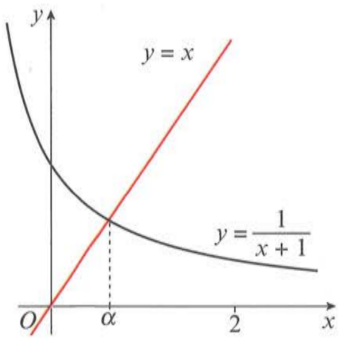
b. Find $\mathrm{f}'(x)$. (3 marks)

c. Show that the x-coordinate of P lies in the interval $[1.8, 1.9]$. (2 marks)

d. Show that the x-coordinate of P is the solution to the equation $x = \arccos\left(\frac{-\sin x}{2x}\right)$. (2 marks)

e. Use the iterative formula $x_{n+1} = \arccos\left(\frac{-\sin x_n}{2x_n}\right)$, $x_0 = 1.8$ to calculate the values of x_1, x_2 and x_3, giving your answers to 3 decimal places. (3 marks)

8. E/P The diagram shows part of the curve with equation $y = \frac{1}{x+1}$



The curve intersects the line $y = x$ at the point where $x = \alpha$.

a. Show that α lies in the interval $\left[\frac{1}{2}, 1\right]$. (2 marks)

In an effort to find a better approximation to α, a student uses the iterative formula

$$x_{n+1} = \frac{1}{x_n + 1}$$

b. Use the graph, together with $x_0 = 2$, to explain whether the iteration will converge or diverge. (2 marks)

10.3 The Newton–Raphson method

Practice Book, printed page 84.

1. $f(x) = x^2 - 6x + 7$

a. Show that the equation $f(x) = 0$ has a root, α, in the interval $[1.5, 1.6]$.

b. Calculate $f'(1.6)$.

c. Work out the value of $1.6 - \dfrac{f(1.6)}{f'(1.6)}$, giving your answer correct to 6 decimal places.

d. Use algebra to find the exact values of the roots of $f(x)$.

e. Comment on the accuracy of your answer from part c.

2. $f(x) = x^4 - 3x^2 + x + 1$

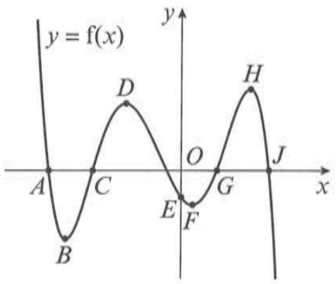
a. Show that the equation $f(x) = 0$ has a root α in the interval $1.2 < \alpha < 1.3$.

b. Using $x_0 = 1.3$ as a first approximation to α, apply the Newton–Raphson procedure once to $f(x)$ to find a second approximation to α, giving your answer to 3 decimal places.

Hint

The Newton–Raphson formula is $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$

3. The diagram shows part of the curve $y = f(x)$.



The x-coordinates of which points would not be valid as a first approximation when applying the Newton–Raphson method to find the roots of $f(x) = 0$?

Hint

If any value, x_i, in the Newton–Raphson method is at a turning point, the method will fail because $f'(x_i) = 0$ and the formula would result in division by zero, which is not valid.

4. E $f(x) = 4\cos^2 x - e^{-x}$

a. Show that the equation $f(x) = 0$ has a root α in the interval $[1.3, 1.4]$. (1 mark)

b. Find $f'(x)$. (2 marks)

c. Using $x_0 = 1.3$ as a first approximation to α, apply the Newton–Raphson procedure once to $f(x)$ to find a second approximation to α, giving your answer to 3 decimal places. (3 marks)

5. E $f(x) = e^{-\frac{1}{4}x} + \dfrac{3}{4}x - \dfrac{1}{4}x^2$

a. Show that the equation $f(x) = 0$ has a root in the interval $[3.4, 3.5]$. (1 mark)

b. Using $x_0 = 3.5$ as a first approximation to α, apply the Newton–Raphson procedure once to $f(x)$ to find a second approximation to α, giving your answer to 3 decimal places. (4 marks)

c. By considering the change of sign of $f(x)$ over an appropriate interval, show that your answer to part b is correct to 3 decimal places. (2 marks)

6. E $f(x) = \dfrac{\cos^2 x}{x} - \dfrac{1}{4}\ln x$

a. Show that $f(x) = 0$ has a root α in the interval $1.2 < \alpha < 1.3$. (2 marks)

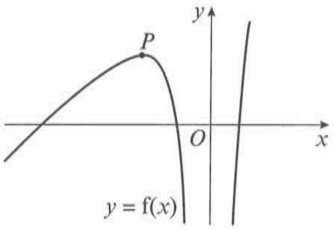
b. Find $f'(x)$. (2 marks)

c. Taking 1.3 as a first approximation to α, apply the Newton–Raphson procedure once to $f(x)$ to

obtain a second approximation for α, giving your answer to 3 decimal places. (3 marks)

7. E/P $\mathrm{f}(x) = x - \dfrac{4}{x^2} + 5$

The diagram shows part of the curve with equation $y = \mathrm{f}(x)$.



The coordinates of the point P are (p, q).

a. Find the value of p and explain clearly why $x_0 = p$ is not suitable to use as a first approximation when applying the Newton–Raphson method. (3 marks)

b. Show that $\mathrm{f}(x) = 0$ has a root α in the interval $[0.8, 0.9]$. (1 mark)

$\mathrm{f}(x) = 0$ also has a root β in the interval $[-5, -4]$.

c. Taking $x_0 = -5$ as a first approximation, apply the Newton–Raphson process once to $\mathrm{f}(x)$ to obtain a second approximation to β. Give your answer to 3 decimal places. (4 marks)

10.4 Applications to modelling

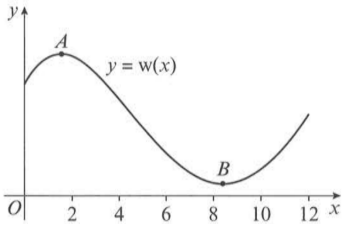
Practice Book, printed page 86.

1. The future world ranking position of a tennis player during a calendar year can be modelled by the function

$$\mathrm{w}(x) = -\frac{1}{50}x^4 + \frac{7}{10}x^3 - 7x^2 + 17x + 40, \quad 0 \leq x \leq 12$$

where x is the number of months since the beginning of the year.

The diagram shows the graph with equation $y = \mathrm{w}(x)$. The graph has a local maximum at A and local minimum at B.



a. Find $\mathrm{w}'(x)$.

b. Show that the player reaches a minimum ranking between 8.3 and 8.4 months after the beginning of the year.

c. Show that the turning points of the graph correspond to the equation

$$x = \pm\sqrt{\frac{10}{21}\left(\frac{2}{25}x^3 + 14x - 17\right)}$$

Hint

The turning points occur when $\mathrm{w}'(x) = 0$ so rearrange your answer from part a.

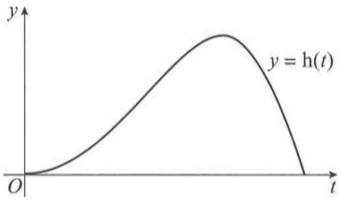
Let $x_0 = 8.3$ and $x_{n+1} = \sqrt{\dfrac{10}{21}\left(\dfrac{2}{25}x_n^3 + 14x_n - 17\right)}$

d. Find the values of x_1, x_2, x_3 and x_4, giving your answers to 3 decimal places.

2. E A boomerang is thrown from a point on level ground. The height in metres of the boomerang above the ground t seconds after it is thrown can be modelled by the function

$$\mathrm{h}(t) = 50\sin\left(\frac{t^2}{36}\right), \quad t \geq 0$$

The diagram shows the sketch of $y = \mathrm{h}(t)$.



a. Show that the boomerang lands between 10 and 11 seconds after it is thrown. (1 mark)

b. Find $\mathrm{h}'(t)$. (2 marks)

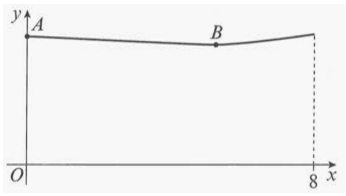
c. Taking 10.5 as a first approximation, apply the Newton–Raphson process once to $\mathrm{h}(t)$ to obtain a second approximation for the time when the boomerang lands. Give your answer to 2 decimal places. (3 marks)

d. By considering the change of sign of $\mathrm{h}'(t)$ in a suitable interval, verify that the boomerang reaches its greatest height 7.52 seconds after it is thrown, correct to 2 decimal places.(2 marks)

3. E/P The trading price, £p, of a stock during a 8-hour trading window can be modelled by the function

$$\mathrm{p}(x) = \frac{1}{3}\cos\left(\frac{x}{3}\right) + \frac{x^2}{100} + 1.29, \quad 0 \leq x \leq 8$$

The diagram shows a sketch of $y = \mathrm{p}(x)$.



a. State the price of the stock, to the nearest penny, when trading begins. (1 mark)

b. Find $\mathrm{p}'(x)$. (2 marks)

c. Show that the price of the stock reaches a minimum at point B between 5 and 5.5 hours after trading opens. (2 marks)

d. Show that the x-coordinate of B is the solution to the equation $x = \frac{50}{9}\sin\left(\frac{x}{3}\right)$. (2 marks)

e. Use the iterative formula $x_{n+1} = \frac{50}{9}\sin\left(\frac{x_n}{3}\right)$, $0 \leq x \leq 8$, with $x_0 = 5.3$, to calculate the values of x_1, x_2 and x_3, giving your answers to 3 decimal places. (3 marks)

4. E/P The value of a mobile phone, t years after purchase, is modelled by the function

$$\mathrm{v}(t) = 450e^{-0.5t} - 40\cos t, \quad t > 0$$

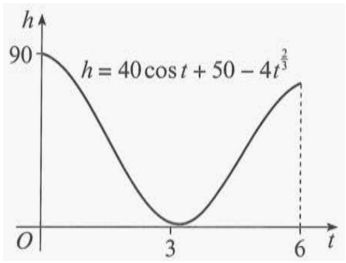
a. Show that $\mathrm{v}(t)$ has a root in the interval $[5, 6]$. (1 mark)

b. Find $\mathrm{v}'(t)$. (2 marks)

c. Taking 5 as a first approximation, apply the Newton–Raphson process once to $\mathrm{v}(t)$ to obtain a second approximation for the time when the value of the mobile phone is zero.
Give your answer to 2 decimal places. (3 marks)

d. Criticise this model with respect to the value of the phone as it gets older. (1 mark)

5. E/P The height of a bungee jumper above the ground can be modelled by the equation $h = 40\cos t + 50 - 4t^{\frac{2}{3}}$, $0 \leq t \leq 6$ where t seconds is the time since the person jumped and h is the height, in metres, above the ground. The bungee jump is considered to be safe if the jumper does not get closer than 10 m to the ground.



a. Show that, some time between 2 and 3 seconds after the person jumps, the bungee jumper is exactly 10 m from the ground. (2 marks)

b. Comment on whether the jump is safe according to this model. (1 mark)

c. Show that the equation $40\cos t + 50 - 4t^{\frac{2}{3}} = 10$ can be written as

$$t = \arccos\left(\frac{4t^{\frac{2}{3}} - 40}{40}\right)$$

(2 marks)

d. Using $t_0 = 2.5$, and the iterative formula $t_{n+1} = \arccos\left(\frac{4t_n^{\frac{2}{3}} - 40}{40}\right)$, find, to 5 decimal places, the values of t_1, t_2, t_3 and t_4. (3 marks)

Problem solving: Set A

Bronze

$f(x) = \frac{1}{2}\cos(x^2) - \frac{1}{2}x + 1$

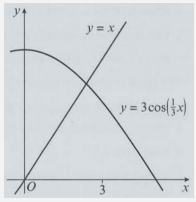
a. Show that there is a root α of $f(x) = 0$ in the interval $[1, 1.5]$. (2 marks)

b. Show that the equation $f(x) = 0$ can be written as $x = \sqrt{\arccos(x-2)}$ (2 marks)

c. Let $x_0 = 1.5$. Use the iterative formula $x_{n+1} = \sqrt{\arccos(x_n - 2)}$ to find, to 3 decimal places, the values of x_1, x_2 and x_3. (3 marks)

Silver

The diagram shows part of the curve with equation $y = 3\cos\left(\frac{1}{3}x\right)$



The curve meets the line $y = x$ at a single point, $x = \alpha$.

a. Show that α is in the interval $[2, 3]$. (2 marks)

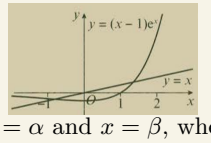
A student uses the iterative formula $x_{n+1} = 3\cos\left(\frac{1}{3}x_n\right)$ in an attempt to find an approximation for α.

b. Using the graph and starting with $x_0 = 3$, explain why the iteration will converge to α. (2 marks)

c. Find x_1, x_2 and x_3, giving your answers to 3 decimal places. (2 marks)

Gold

The diagram shows part of the curve with equation $y = (x-1)e^x$



The curve meets the line $y = x$ at a two points, $x = \alpha$ and $x = \beta$, where $\alpha < 0$ and $\beta > 0$.

a. Show that β is in the interval $1 < \beta < 2$. (2 marks)

A student uses the iterative formula $x_{n+1} = (x_n - 1)e^{x_n}$ in an attempt to find an approximation for β.

b. Using the graph, describe what will happen for the following starting values.

i. $x_0 = \frac{1}{2}$

ii. $x_0 = 2$ (4 marks)

c. By choosing a suitable interval, show that $\alpha = -0.806$ correct to 3 decimal places. (2 marks)

Problem solving: Set B

Bronze

$f(x) = -\frac{4}{3}x + 3x^{\frac{4}{3}} - 2$

a. Show that there is a root α of $f(x) = 0$ in the interval $[1.1, 1.2]$. (2 marks)

b. Taking $x_0 = 1.2$ as a first approximation to α, apply the Newton–Raphson process once to $f(x)$ to obtain a second approximation to α. Give your answer to 3 decimal places. (2 marks)

c. By considering a change of sign of $f(x)$ in a suitable interval, verify that your answer to part b is correct to 2 decimal places. (2 marks)

Silver

$f(x) = e^{-x}\sin(x^2)$

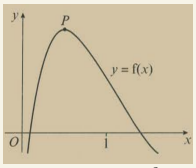
a. Show that there is a root α of $f(x) = 0$ in the interval $1.5 < \alpha < 2$. (2 marks)

b. Taking $x_0 = 1.75$ as a first approximation to α, apply the Newton–Raphson process once to $f(x)$ to obtain a second approximation to α. Give your answer to 3 decimal places. (2 marks)

c. By considering a change of sign of $f(x)$ in a suitable interval, show that your answer to part b is correct to 3 decimal places. (2 marks)

Gold

The diagram shows the curve with equation $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = 12xe^{-2x} - 1$.



The point P, with coordinates (p, q), is a stationary point on the curve. The equation $\mathrm{f}(x) = 0$ has a root α in the interval $1 < \alpha < 2$.

a. Find the coordinates of the point P, leaving your answer in terms of e where necessary. (3 marks)

b. Explain why $x_0 = p$ is not suitable to use as a first approximation if using the Newton–Raphson process to find an approximation of α. (1 mark)

c. Taking $x_0 = 1.5$ as a first approximation to α, apply the Newton–Raphson process once to $\mathrm{f}(x)$ to obtain a second approximation to α. Give your answer to 2 decimal places. (2 marks)

Now try this ⟶ Exam question bank Q2, Q12, Q18, Q22, Q29, Q37

Selected exam question bank questions

Question 2

$\mathrm{f}(x) = \dfrac{1}{3x-7} - 1$

a. Calculate f(2) and f(2.5). (2 marks)

A student writes:

> f(x) changes sign in the interval $[2, 2.5]$ so the equation f(x) = 0 must have a root in this interval.

b. Explain why the student is incorrect. (2 marks)

Question 12

$\mathrm{p}(x) = \dfrac{1}{3}x^2 - 3\sin x + 1$. The equation p($x$) = 0 has a root α between 0 and 1.

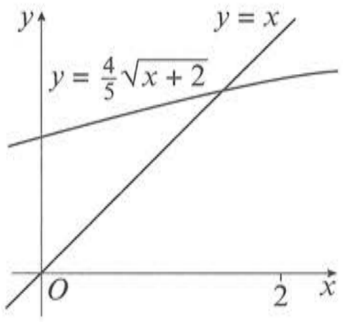
a. Show that the equation p(x) = 0 can be written as $x = \arcsin\left(\dfrac{1}{9}x^2 + \dfrac{1}{3}\right)$. (2 marks)

b. Use the iterative formula $x_{n+1} = \arcsin\left(\dfrac{1}{9}x_n^2 + \dfrac{1}{3}\right)$ with $x_0 = 0.5$ to obtain x_1, x_2 and x_3. (2 marks)

c. Taking 0.4 as a first approximation to α, apply the Newton–Raphson procedure once to obtain a second approximation to α, giving your answer to 3 decimal places. (3 marks)

Question 18

The diagram shows part of the curve with equation $y = \dfrac{4}{5}\sqrt{x+2}$



The curve intersects the line $y = x$ at the point $x = \alpha$.

a. Show that $1 < \alpha < 2$. (3 marks)

A student uses the iterative formula $x_{n+1} = \dfrac{4}{5}\sqrt{x_n + 2}$ together with $x_0 = 1$ to find a better approximation for α.

b. With reference to the graph above, determine whether or not this iterative formula can be used to find an approximation for α, justifying your answer. (2 marks)

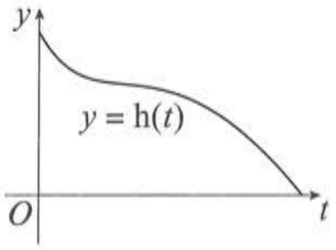
Question 22

The height above ground of a person on a water slide can be modelled by the equation

$$\mathrm{h}(t) = 18e^{-0.5t} + 10\sin\left(\frac{t}{3}\right) - \frac{1}{10}t, \quad t \geq 0$$

where t is the time in seconds since the person began sliding.

The diagram shows the graph of $y = \mathrm{h}(t)$.



a. Show that the person reaches the bottom of the slide between 9 and 10 seconds after they start sliding. (1 mark)

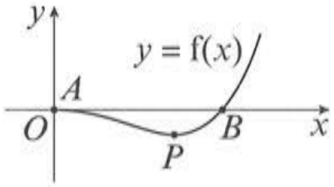
b. Taking 9 as a first approximation, apply the Newton–Raphson procedure once to $\mathrm{h}(t)$ to obtain a second approximation for the time when the person reaches the bottom of the slide, giving your answer to 3 decimal places. (3 marks)

c. By considering the change of sign of $\mathrm{h}(t)$ in a suitable interval, verify that your answer to part b is in fact correct to 3 decimal places. (2 marks)

Question 29

The diagram shows the part of the graph of $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = x^3 \ln x$.

The curve cuts the x-axis at A and B and has a turning point at P.



a. Work out the coordinates of A and B. (1 mark)

b. Find $\mathrm{f}'(x)$. (3 marks)

c. By considering the change of sign of $\mathrm{f}'(x)$ in a suitable interval, verify that the x-coordinate of P is 0.717 correct to 3 decimal places. (2 marks)

d. Find the exact value of the x-coordinate of P. (2 marks)

Question 37

a. On the same set of axes, sketch the curve with equation $y = 2\ln x$ and the line with equation $2x + 5y = 10$. (2 marks)

b. With reference to your sketch, explain why the function $\mathrm{f}(x) = 10\ln x + 2x - 10$ has exactly one root. (2 marks)

c. Show that $\mathrm{f}(x)$ has a root between $x = 1.8$ and $x = 1.9$. (2 marks)

Answers

10.1 Locating roots

1.

a. The graph crosses the x-axis between $x = -1$ and $x = 0$.

b.

i. -0.464

ii. 0.574

c. There is a change of sign between $x = 2.2$ and $x = 2.3$.

2.

a. $f(-1.2) = -0.176 < 0$, $f(-1.1) = 0.733 > 0$
Sign change implies at least one root in interval.

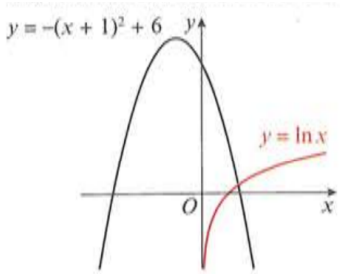
b. $f(0.6) = 0.1718\ldots > 0$, $f(0.7) = -0.2022\ldots < 0$
Sign change implies at least one root in interval.

c. $f(-1.11) = 0.0403\ldots > 0$, $f(-1.10) = -0.1265\ldots < 0$
Sign change implies at least one root in interval.

d. $f(2.37) = 0.0011\ldots > 0$, $f(2.38) = -0.0140\ldots < 0$
Sign change implies at least one root in interval.

3.

a.



b. If $\ln x = -(x+1)^2 + 6$, then $\ln x + (x+1)^2 - 6 = 0$.
As the graphs of $y = \ln x$ and $y = -(x+1)^2 + 6$ only intersect once, $f(x)$ has only one root.

c. $f(1.3) = -0.4476\ldots < 0$, $f(1.4) = 0.0964\ldots > 0$
Sign change implies at least one root in interval.

4.

a. $-2\cos x \sin x + 2e^{-2x}$

b. $f'(3.1) = 0.0871\ldots > 0$, $f'(3.2) = -0.1132\ldots < 0$
As $f'(x)$ changes sign in the interval $[3.1, 3.2]$, there must be a stationary point in this interval.

c. $f'(3.1425) = 0.0019\ldots > 0$, $f'(3.1435) = -0.00009\ldots < 0$
There is a sign change in the interval $(3.1425, 3.1435)$ so $3.1425 < \alpha < 3.1435$, so $\alpha = 3.143$ to 3 d.p.

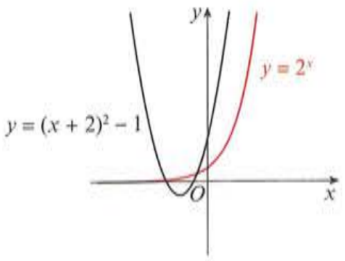
5.

a. $g(1.1) = -0.2416\ldots < 0$, $g(1.2) = 0.1392\ldots > 0$
Sign change implies at least one root in interval.

b. $g(1.1645) = -0.0008\ldots < 0$, $g(1.1655) = 0.0030\ldots > 0$
There is a sign change in the interval $(1.1645, 1.1655)$ so $1.1645 < \alpha < 1.1655$, so $\alpha = 3.165$ to 3 d.p.

6.

a.



b. If $(x+2)^2 - 1 = 2^x$ then $2^x + 1 - (x+2)^2 = 0$
As $y = (x+2)^2 - 1$ and $y = 2^x$ intersect twice, $\mathrm{f}(x) = 2^x + 1 - (x+2)^2$ has two roots.

c. $\mathrm{f}(-0.8) = 0.1343\ldots > 0$, $\mathrm{f}(-0.7) = -0.0744\ldots < 0$
Sign change implies at least one root in interval.

7.

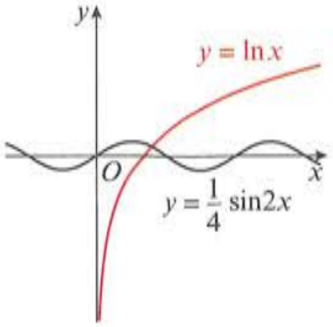
a. $\mathrm{f}(1.9) = 2.67$ (2 d.p.), $\mathrm{f}(2.1) = -10.67$ (2 d.p.)

b. $\mathrm{f}(2)$ is undefined.

c. $\dfrac{11}{6}$

8.

a.



b. If $\frac{1}{4}\sin 2x = \ln x$ then $\sin 2x - 4\ln x = 0$. As $y = \frac{1}{4}\sin 2x$ and $y = \ln x$ only intersect once, $\mathrm{h}(x) = \sin 2x - 4\ln x$ has only one root.

c. $\mathrm{h}(1.18) = 0.0423\ldots > 0$, $\mathrm{h}(1.19) = -0.0057\ldots < 0$
Sign change implies at least one root in interval.

10.2 Iteration

1.

a. $x^2 = 8x - 3 \Rightarrow x = \sqrt{8x-3}$

b. Divide by x: $x - 8 + \dfrac{3}{x} = 0 \Rightarrow x = 8 - \dfrac{3}{x}$, $x \neq 0$

c. $8x = x^2 - 3 \Rightarrow x = \dfrac{x^2-3}{8}$

2.

a. $x^2 = 4x + 2 \Rightarrow x = \sqrt{4x+2}$

b. $x_1 = 4.690$, $x_2 = 4.556$, $x_3 = 4.497$, $x_4 = 4.471$

3.

a.

i. $x^3 = 3x^2 - 1 \Rightarrow x = \sqrt[3]{3x^2-1}$

ii. $3x^2 = x^3 + 1 \Rightarrow x = \sqrt{\dfrac{x^3+1}{3}}$

b. $x_0 = 1.000$, $x_1 = 1.260$, $x_2 = 1.555$, $x_3 = 1.843$, $x_4 = 2.094$

c. $x_1 = 0.816$, $x_2 = 0.717$, $x_3 = 0.676$, $x_4 = 0.660$

d. The iterations appear to be converging to different numbers.

4.

a. $\frac{1}{2}x^2 = 2\sin x \Rightarrow x = \sqrt{4\sin x}$

b. $x_1 = 1.997$, $x_2 = 1.908$, $x_3 = 1.943$

c. $\text{f}(1.9335) = 0.0006\ldots > 0$, $\text{f}(1.9345) = -0.0019\ldots < 0$
Sign change implies at least one root in interval, so $\alpha = 1.934$ to 3 d.p.

5.

a. $A\left(\frac{1-\sqrt{5}}{2}, 0\right)$, $C\left(\frac{1+\sqrt{5}}{2}, 0\right)$

b. $5x^2 = x^5 + 2 \Rightarrow x = \sqrt{\frac{x^5+2}{5}}$

c. $x_1 = 0.637$, $x_2 = 0.649$, $x_3 = 0.650$

d. $\text{f}(0.6505) = 0.0007\ldots > 0$, $\text{f}(0.6515) = -0.0048\ldots < 0$
Sign change implies at least one root in interval, so $\alpha = 0.651$ to 3 d.p.

6.

a. $\text{f}(-1.9) = -0.0288\ldots < 0$, $\text{f}(-1.8) = 0.6450\ldots > 0$
Sign change implies at least one root in interval.

b. $e^{x+4} = 10 - \frac{1}{2}x^2 \Rightarrow x + 4 = \ln\left(10 - \frac{1}{2}x^2\right)$
$\Rightarrow x = \ln\left(10 - \frac{1}{2}x^2\right) - 4$

c. $x_1 = -2.295$, $x_2 = -2.003$, $x_3 = -1.921$

d. $\text{f}(-1.89545) = -0.0002\ldots < 0$, $\text{f}(-1.89535) = 0.0004\ldots > 0$
Sign change implies at least one root in interval, so $\alpha = -1.8954$ to 4 d.p.

7.

a. Substitute $x = 0$ and $x = \pi$ into $\text{f}(x)$.

b. $\sin^2 x + 2x\sin x\cos x$

c. $\text{f}'(1.8) = 0.1518\ldots > 0$, $\text{f}'(1.9) = -0.2670\ldots < 0$
Sign change implies at least one turning point in interval.

d. $\sin^2 x + 2x\sin x\cos x = 0 \Rightarrow 2x\sin x\cos x = -\sin^2 x$
$\Rightarrow \cos x = \frac{-\sin x}{2x} \Rightarrow x = \arccos\left(\frac{-\sin x}{2x}\right)$

e. $x_1 = 1.845$, $x_2 = 1.835$, $x_3 = 1.837$

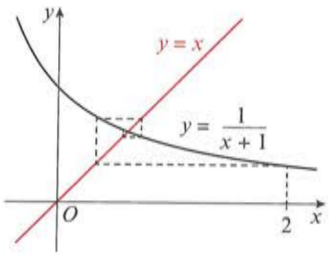
8.

a. Let $f(x) = \dfrac{1}{x+1} - x$

$f\left(\dfrac{1}{2}\right) = 0.1666\ldots > 0$, $f(1) = -0.5 < 0$

Sign change implies at least one root in interval.

b.



The iteration will converge.

10.3 The Newton–Raphson method

1.

a. $f(1.5) = 0.25 > 0$, $f(1.6) = -0.04 < 0$

Sign change implies at least one root in interval.

b. -2.8

c. 1.585714

d. $3 - \sqrt{2}$, $3 + \sqrt{2}$

e. $3 - \sqrt{2} \approx 1.585786$, so part c answer correct to 3 d.p.

2.

a. $f(1.2) = -0.0464\ldots < 0$, $f(1.3) = 0.0861\ldots > 0$

Sign change implies at least one root in interval.

b. 1.257

3. B, D, F and H

4.

a. $f(1.3) = 0.0136\ldots > 0$, $f(1.4) = -0.1310\ldots < 0$

Sign change implies at least one root in interval.

b. $-8\sin x \cos x + e^{-x}$

c. 1.308

5.

a. $f(3.4) = 0.0874\ldots > 0$, $f(3.5) = -0.0206\ldots < 0$

Sign change implies at least one root in interval.

b. 3.481

c. $f(3.4805) = 0.0008\ldots > 0$, $f(3.4815) = -0.0002\ldots < 0$

Sign change implies at least one root in interval, so $\alpha = 3.481$ to 3 d.p.

6.

a. $f(1.2) = 0.0638\ldots > 0$, $f(1.3) = -0.0105\ldots < 0$

Sign change implies at least one root in interval.

b. $f'(x) = \dfrac{-2x\sin x\cos x - \cos^2 x}{x^2} - \dfrac{1}{4x}$

c. 1.283

7.

a. $p = -2$. Gradient at point P is zero, so using Newton–Raphson formula would result in division by zero.

b. $\mathrm{f}(0.8) = -0.45 < 0$, $\mathrm{f}(0.9) = 0.96 > 0$
Sign change implies at least one root in interval.

c. -4.829

10.4 Applications to modelling

1.

a. $-\frac{2}{25}x^3 + \frac{21}{10}x^2 - 14x + 17$

b. $\mathrm{w}'(8.3) = -0.27396\ldots < 0$, $\mathrm{w}'(8.4) = 0.15968\ldots > 0$
There is a change of sign in the interval $8.3 < x < 8.4$, so there is at least one turning point in this interval.

c. $-\frac{2}{25}x^3 + \frac{21}{10}x^2 - 14x + 17 = 0$
$\Rightarrow \frac{21}{10}x^2 = \frac{2}{25}x^3 + 14x - 17$
$\Rightarrow x^2 = \frac{10}{21}\left(\frac{2}{25}x^3 + 14x - 17\right)$
$\Rightarrow x = \pm\sqrt{\frac{10}{21}\left(\frac{2}{25}x^3 + 14x - 17\right)}$

d. $x_0 = 8.3$, $x_1 = 8.308$, $x_2 = 8.315$, $x_3 = 8.321$, $x_4 = 8.326$

2.

a. $\mathrm{h}(10) = 17.7920\ldots > 0$, $\mathrm{h}(11) = -10.8879\ldots < 0$
Sign change implies at least one root in interval, therefore the boomerang lands between 10 and 11 seconds.

b. $\frac{25t}{9}\cos\left(\frac{t^2}{36}\right)$

c. 10.64 seconds

d. $\mathrm{h}'(7.515) = 0.0425\ldots > 0$, $\mathrm{h}'(7.525) = -0.0446\ldots < 0$
Sign change implies at least one turning point in interval, so boomerang is at a maximum height at 7.52 seconds, correct to 2 decimal places.

3.

a. £1.62

b. $-\frac{1}{9}\sin\left(\frac{x}{3}\right) + \frac{x}{50}$

c. $\mathrm{p}'(5) = -0.0106\ldots < 0$, $\mathrm{p}'(5.5) = 0.0026\ldots > 0$
Sign change implies at least one turning point in interval.

d. $-\frac{1}{9}\sin\left(\frac{x}{3}\right) + \frac{x}{50} = 0 \Rightarrow \frac{x}{50} = \frac{1}{9}\sin\left(\frac{x}{3}\right) \Rightarrow x = \frac{50}{9}\sin\left(\frac{x}{3}\right)$

e. $x_1 = 5.449$, $x_2 = 5.389$, $x_3 = 5.415$

4.

a. $\mathrm{v}(5) = 25.592\ldots > 0$, $\mathrm{v}(6) = -16.003\ldots < 0$
Sign change implies at least one root in interval.

b. $-225e^{-0.5t} + 40\sin t$

c. 5.45 years

d. $\mathrm{v}(6) \approx -16$, and the value cannot be negative, so this model is not suitable as the phone gets older than 6 years.

5.

a. $t = 2 \Rightarrow h = 27.0045\ldots > 10$, $t = 3 \Rightarrow h = 2.0799\ldots < 10$.
Therefore at some time between $t = 2$ and $t = 3$ the person is exactly 10 m from the ground.

b. At $t = 3$ the person is approximately 2.1 m from the ground, so the jump is not safe.

c. $40\cos t + 50 - 4t^{\frac{2}{3}} = 10 \Rightarrow 40\cos t = 4t^{\frac{2}{3}} - 40$
$\Rightarrow \cos t = \dfrac{4t^{\frac{2}{3}} - 40}{40} \Rightarrow t = \arccos\left(\dfrac{4t^{\frac{2}{3}} - 40}{40}\right)$

d. $t_1 = 2.52490$, $t_2 = 2.52280$, $t_3 = 2.52297$, $t_4 = 2.52296$

Problem solving: Set A

Bronze

a. $\mathrm{f}(1) = 0.7701\ldots > 0$, $\mathrm{f}(1.5) = -0.0640\ldots < 0$
Sign change implies at least one root in interval.

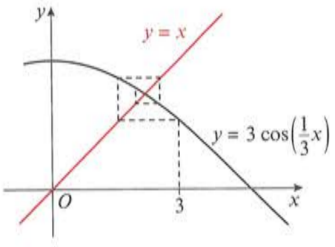
b. $\dfrac{1}{2}\cos(x^2) = \dfrac{1}{2}x - 1 \Rightarrow \cos(x^2) = x - 2$
$\Rightarrow x^2 = \arccos(x - 2) \Rightarrow x = \sqrt{\arccos(x - 2)}$

c. $x_1 = 1.447$, $x_2 = 1.469$, $x_3 = 1.460$

Silver

a. Let $\mathrm{f}(x) = x - 3\cos\left(\dfrac{1}{3}x\right)$
$\mathrm{f}(2) = -0.3576\ldots < 0$, $\mathrm{f}(3) = 1.3790\ldots > 0$
Sign change implies at least one root in interval.

b.



c. $x_1 = 1.621$, $x_2 = 2.573$, $x_3 = 1.963$

Gold

a. Let $\mathrm{f}(x) = x - (x - 1)e^x$. $\mathrm{f}(1) = 1 > 0$, $\mathrm{f}(2) = -5.389\ldots < 0$
Sign change implies at least one root in interval.

b.

i. The iteration will converge to the other root, α.

ii. The iteration will diverge.

c. $\mathrm{f}(-0.8065) = -0.00004\ldots < 0$, $\mathrm{f}(-0.8055) = 0.0013 > 0$
Sign change implies at least one root in interval, so $\alpha = -0.806$ to 3 d.p.

Problem solving: Set B

Bronze

a. $\mathrm{f}(1.1) = -0.0601\ldots < 0$, $\mathrm{f}(1.2) = 0.2257\ldots > 0$
Sign change implies at least one root in interval.

b. 1.123

c. $\mathrm{f}(1.1225) = -0.003\ldots < 0$, $\mathrm{f}(1.1235) = 0.0005\ldots > 0$
Sign change implies at least one root in interval, so part b answer is correct to 3 d.p.

Silver

a. $f(1.5) = 0.1736\ldots > 0$, $f(2) = -0.1024\ldots < 0$
Sign change implies at least one root in interval.

b. 1.772

c. $f(1.7715) = 0.0005\ldots > 0$, $f(1.7725) = -0.00002\ldots < 0$
Sign change implies at least one root in interval, so part b answer is correct to 3 d.p.

Gold

a. $\left(\frac{1}{2}, \frac{6}{e} - 1\right)$

b. $f'(p) = 0$, so would mean dividing by zero in the Newton–Raphson formula, which is not valid.

c. 1.41

Selected exam question bank answers

Question 2

a. $f(2) = -2$, $f(2.5) = 1$

b. $x = \frac{7}{3}$ is an asymptote and as $2 < \frac{7}{3} < 2.5$, a change of sign in this interval will be caused by the asymptote, so there doesn't have to be a root in this interval.

Question 12

a. $\frac{1}{3}x^2 - 3\sin x + 1 = 0 \Rightarrow 3\sin x = \frac{1}{3}x^2 + 1$
$\Rightarrow x = \arcsin\left(\frac{1}{9}x^2 + \frac{1}{3}\right)$

b. $x_1 = 0.369$, $x_2 = 0.356$, $x_3 = 0.355$ (3 d.p.)

c. $\alpha = 0.354$

Question 18

a. Let $f(x) = \frac{4}{5}\sqrt{x+2} - x$
$f(1) = 0.3856\ldots > 0$, $f(2) = -0.4 < 0$
Sign change implies at least one root in interval.

b. Yes, the iteration will converge towards α.

Question 22

a. $h(9) = 0.7111\ldots > 0$, $h(10) = -2.7843\ldots < 0$
Sign change implies at least one root in interval, so person reaches bottom of slide between 9 and 10 seconds.

b. 9.203

c. $h(9.2025) = 0.0007\ldots > 0$, $h(9.2035) = -0.0028\ldots < 0$
Sign change implies at least one root in interval, so part b answer is correct to 3 d.p.

Question 29

a. $A(0, 0)$ and $B(1, 0)$

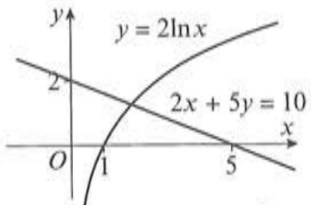
b. $3x^2 \ln x + x^2$

c. $f'(0.7165) = -0.00006\ldots < 0$, $f'(0.7175) = 0.0020\ldots > 0$
Sign change implies at least one turning point in interval, so x-coordinate of P is 0.717 to 3 d.p.

d. $e^{-\frac{1}{3}}$

Question 37

a.



b. $2x + 5y = 10 \Rightarrow y = 2 - \frac{2}{5}x$

$2\ln x = 2 - \frac{2}{5}x \Rightarrow 2\ln x + \frac{2}{5}x - 2 = 0$

$\Rightarrow 10\ln x + 2x - 10 = 0$

As the curve $y = 2\ln x$ and the line $2x + 5y = 10$ only intersect once, f(x) has exactly one root.

c. f(1.8) $= -0.5221\ldots < 0$, f(1.9) $= 0.2185\ldots > 0$

Sign change implies the root is in this interval.